(I) find the complementary function for the following equations

(1) $(D^3 - 3D^2 + 2D)y = 0$; Ans:- $y = C_1 e^{0x} + C_2 e^{x} + C_3 e^{2x}$

(2) $(D^3 - 8)y = 0$; Ans:- $y = C_1 e^{2x} + e^{-x}\{C_2 \cos(x\sqrt{3}) + C_3 \sin(\sqrt{3}x)\}$

(3) $(D^2 + D + 1)^2 y = 0$; Ans:- $y = e^{-x/2}\left[(C_1 + C_2 x)\cos(x\sqrt{3}/2) + (C_3 + C_4 x)\sin\left(\frac{x\sqrt{3}}{2}\right)\right]$

(4) $[(D^2+1)^2 (D^2+D+1)]y = 0$; Ans:- $y = (C_1 + C_2 x)\cos x + (C_3 + C_4 x)\sin x$

(5) ~~$(D^4 + 3D^3 + 3D^2 + 2D + 1)y = 0$; Ans:- $y =$~~

(5) $\frac{d^2y}{dx^2} + 4y = 0$; Ans:- $y = C_1 \cos 2x + C_2 \sin 2x$

(2) ~~Find~~ Determine the particular integral for the following eqns

(1) $(D^2 - 3D + 2)y = e^{3x}$; Ans: $\frac{e^{x}}{36}$

(2) $(D^2 - 4D + 4)y = e^{2x} - e^{-2x}$; Ans: $\frac{1}{16} e^{2x} - \frac{x^2}{2} e^{-2x}$

(3) $(D^3 - 1)y = (e^x + 1)^2$; Ans: $\frac{1}{7} e^{2x} + \frac{2x}{3} e^{x} - 1$

(4) $(D^2 - 2kD + k^2)y = e^{3x}$; Ans: $\frac{x^2}{2} e^{kx}$

(5) $(D^2 - a^2)y = \cosh ax$; Ans: $\left(\frac{x}{2a}\right)\sinh ax$

(3) Determine the P.I for the following equations

(1) $(D^2 + 1)y = \cos 2x$; Ans: $-\frac{1}{3}\cos 2x$

(2) $(D-1)^2 (D^2+1)^2 y = \sin x$; Ans: $\frac{x}{4}\sin x$

(3) $(D^4 - m^4)y = \sin mx$; Ans: $\frac{x}{4m^3}\cos mx$

(4) $(D^2 + 4)y = \sin^2 x$; Ans: $\frac{1}{8} - \frac{x}{8}\sin 2x$

(5) $(D^2 + 4)y = a \sin x \cos x$; Ans: $-\frac{ax}{8}\cos 2x$

(4) Determine the perticular integral for the following eqns

(1) $(D^2+D)y = x^2+2x+4$; Ans: $\frac{x^3}{3}+4x$

(2) $(D^4-a^4)y = x^4$; Ans: $-\frac{1}{a^4}\left(x^4+\frac{24}{a^4}\right)$

(3) $(D^3-D^2-6D)y = x^2+1$; Ans: $-\frac{1}{18}\left(x^3-\frac{x^2}{2}+\frac{25x}{6}\right)$

(4) $(D^4-2D^3+5D^2-8D+4)y = x^2$; Ans: $\frac{1}{4}\left(x^2+4x+\frac{11}{2}\right)$

(5) $(D^2-1)y = 0$; Ans: -1

(5) Determine the perticular integral for the following eqns

(1) $(D^2-2D+1)y = x^3e^{3x}$; Ans: $\frac{1}{8}e^{3x}(2x^2-4x+3)$

(2) $(D-1)^2y = e^x\sec^2x\tan x$; Ans: $\frac{e^x}{2}(\tan x - x)$

(3) $(D^2+3D+2)y = e^{2x}\sin x$; Ans: $\frac{1}{170}e^{2x}(11\sin x - 7\cos x)$

(4) $(D^3-D^2+3D+5)y = e^x\cos x$; Ans: $\frac{1}{34}e^x(3\sin x+5\cos x)$

(5) $(D^2-1)y = \cosh x\cos x$; Ans: $\frac{2}{5}\sin x\sinh x - \frac{1}{5}\cos x\cosh x$

(6) $(D^2-5D+6)y = e^{4x}(x^2+9)$; Ans: $\frac{1}{4}e^{4x}(2x^2-6x+25)$

(6) Determine the perticular integral for the following eq

(1) $(D^2+2D+1)y = x\sin x$; Ans: $\frac{1}{2}((x-1)\sin x+\cos x)$

(2) $(D^4+2D^2+1)y = x^2\cos x$; Ans: $-\frac{1}{4}\left[(x^4/12-3x^2/4)\cos x-(x^3/3)\sin x\right]$

(7)

(3) $(D^4-1)y = xe^x\sin x$.